

Java Core Programming Guide

Complete 10-Module OOP & Reference Notes

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Module 1: Java Basics & JVM Architecture

MODULE 1

1. Java Kya Hai?

Java ek class-based, object-oriented programming language hai jo "Write Once, Run Anywhere" (WORA) principle par kaam karti hai.

```
public class Main {
    public static void main(String[] args) {
        System.out.println("Namaste World! Welcome to Java.");
    }
}
```

1. Primitive Data Types

Primitive types: `int` , `double` , `float` , `char` , `boolean` , `byte` , `short` , `long` .

```
int age = 20;
double salary = 45000.50;
boolean isEligible = true;
```

Module 3: Control Flow Statements

1. Decision Making & Loops

```
int mark = 80;
if (mark >= 75) {
    System.out.println("Distinction");
} else {
    System.out.println("Pass");
}

for (int i = 1; i <= 5; i++) {
    System.out.println("Iteration: " + i);
}
```

Module 4: Arrays & Strings in Java

1. Array & String Class

```
int[] numbers = {10, 20, 30, 40};
String name = "Java Developer";
System.out.println("Length: " + name.length());
```

1. Class & Getter/Setters

```
class Student {  
    private String name;  
  
    public String getName() { return name; }  
    public void setName(String name) { this.name = name; }  
}
```

1. Extending Classes & Method Overriding

```
class Animal {  
    void makeSound() { System.out.println("Animal sound"); }  
}  
  
class Dog extends Animal {  
    @Override  
    void makeSound() { System.out.println("Bark!"); }  
}
```

1. Abstract Classes vs Interfaces

```
interface Drawable {  
    void draw();  
}  
  
class Circle implements Drawable {  
    public void draw() { System.out.println("Drawing Circle"); }  
}
```

1. Try-Catch-Finally Blocks

```
try {
    int data = 50 / 0;
} catch (ArithmeticException e) {
    System.out.println("Cannot divide by zero: " + e);
} finally {
    System.out.println("Always executed block.");
}
```

Module 9: Java Collections Framework

1. ArrayList, HashSet & HashMap

```
import java.util.*;

List list = new ArrayList<>();
list.add("Java");
list.add("Python");

Map map = new HashMap<>();
map.put("Rahul", 101);
```

Module 10: Multithreading & File I/O

1. Creating Threads & File Operations

```
class MyThread extends Thread {
    public void run() {
        System.out.println("Thread is running...");
    }
}

// Usage: MyThread t1 = new MyThread(); t1.start();
```